HOMEWORK 8.2 · UNIT 8 · NC.8.G.3 · NC.8.G.2

Transformations Day 2

Targets: G.3-3, G.2-2, G.3-6, G.3-7, G.2-6, G.2-8, G.2-7

Name _____ Date _____ Block _____

Remember: Rotations about the origin turn **counterclockwise** unless it says clockwise. **90°**: $(x, y) \to (-y, x)$. **180°**: $(x, y) \to (-x, -y)$. **270°** (same as 90° clockwise): $(x, y) \to (y, -x)$. Rotations are rigid. **Example:** $(3, 2)$ rotated $90° \to (-2, 3)$.

SCORE 3.0 Approaching · the basics

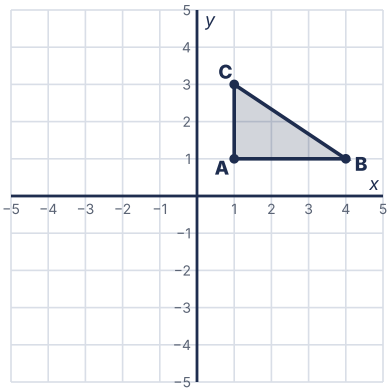
Targets: G.3-3, G.2-2

1 Rotate U(4, −2) 90° counterclockwise about the origin. Find U′. G.3-3

Ⓐ (2, 4) Ⓑ (−2, −4) Ⓒ (−4, 2) Ⓓ (−2, 4)

2 Rotate R(−2, 2) 180° about the origin. Find R′. G.3-3

3 Rotate △ABC 90° counterclockwise about the origin. A(1, 1), B(4, 1), C(1, 3). Write A′, B′, C′. Draw the image. G.2-2



4 Reflect M(−3, 5) over the y-axis. Find M′. REVIEW 8.1 G.3-3

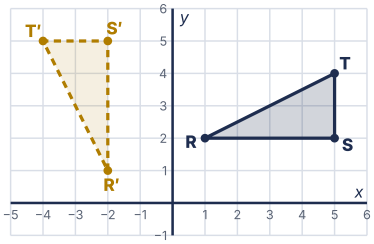
Ⓐ (−3, −5) Ⓑ (3, 5)
Ⓒ (3, −5) Ⓓ (5, −3)

5 Reflecting a point over the y-axis changes the sign of its y-coordinate. REVIEW 8.1 G.3-3

True False

SCORE 4.0 Proficient · on target Targets: G.3-6, G.3-7, G.2-6

6 Which transformation maps △RST onto △R′S′T′? G.3-6

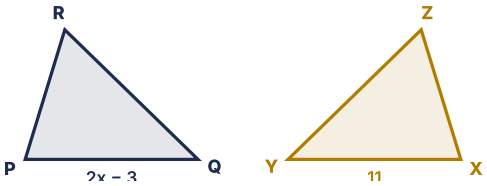


(A) Rotation 90° clockwise
(B) Rotation 90° counterclockwise
(C) Rotation 180°
(D) Reflection over the *y*-axis

7 **Dev** made a mistake. Circle it. Explain it. Fix it. G.3-7

DEV'S WORK
Rotate (−4, −1) 90° clockwise.
Image = (1, −4)

8 △PQR ≅ △XYZ. Find *x*. Then find PQ. G.2-6



Hint: PQ matches XY.

SCORE 5.0 Mastery · go deeper Targets: G.2-8, G.2-7

9 Jae rotates a figure 90° clockwise three times. What single rotation does the same thing? Why is one step better? G.2-8

It is the same as _____. One step is better because _____.

10 △ABC has A(1, 2), B(3, 2), C(1, 5). Its final image is A″(2, −2), B″(0, −2), C″(2, −5). Describe a sequence that does this. Are the triangles congruent? Explain. G.2-7

First _____. Then _____. They _____ congruent because _____.